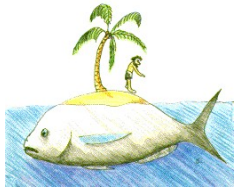
eccw05
17-21 April 2023 · Bergen, Norway

Can a spatial trophic ecosystem model represent observed changes in functional groups distribution due to environmental changes in the Barents Sea Large Marine Ecosystem?

Marcela C. Nascimento, Bérengère Husson, Lilia Guillet, Torstein Pedersen

Main questions

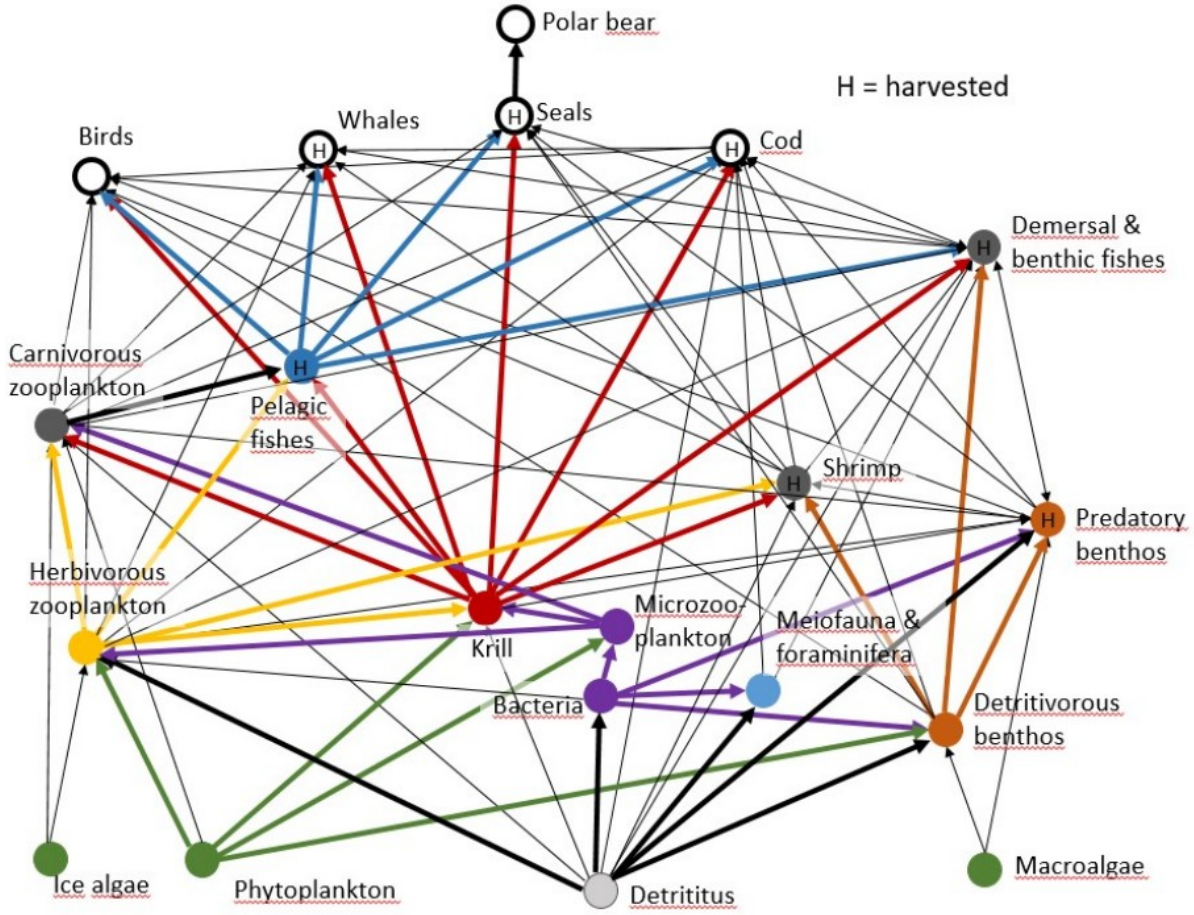


Q1 How well can a spatial dynamic model predict spatial distributions of functional groups in the last two decades?

Q2 Can the model predict the changes in observed spatial distribution of the functional groups from cold to warm years?

frontiers in Marine Science

ORIGINAL RESEARCH
published: 17 September 2021
doi: 10.3389/fmars.2021.732637



Overexploitation, Recovery, and Warming of the Barents Sea Ecosystem During 1950–2013

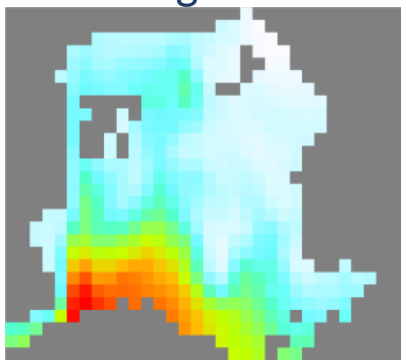
Torstein Pedersen[1], Nina Mikkelsen[1,2], Ulf Lindstrøm[1,2], Paul E. Renaud[3,4], Marcela C. Nascimento[1], Marie-Anne Blanchet[5†], Ingrid H. Ellingsen[6], Lis L. Jørgensen[2] and Hugues Blanchet[3,7]*

High resolution mass balanced model with 108 Functional groups (FG)

Trophic flows

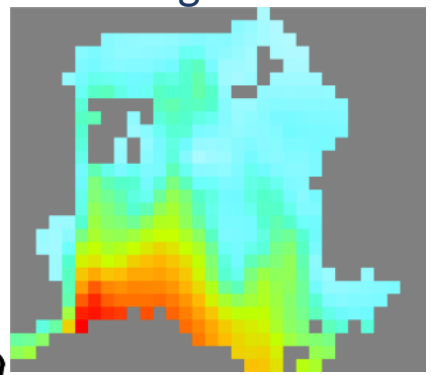
2004 Bottom T°C

Average 0.62 C



2013 Bottom T°C

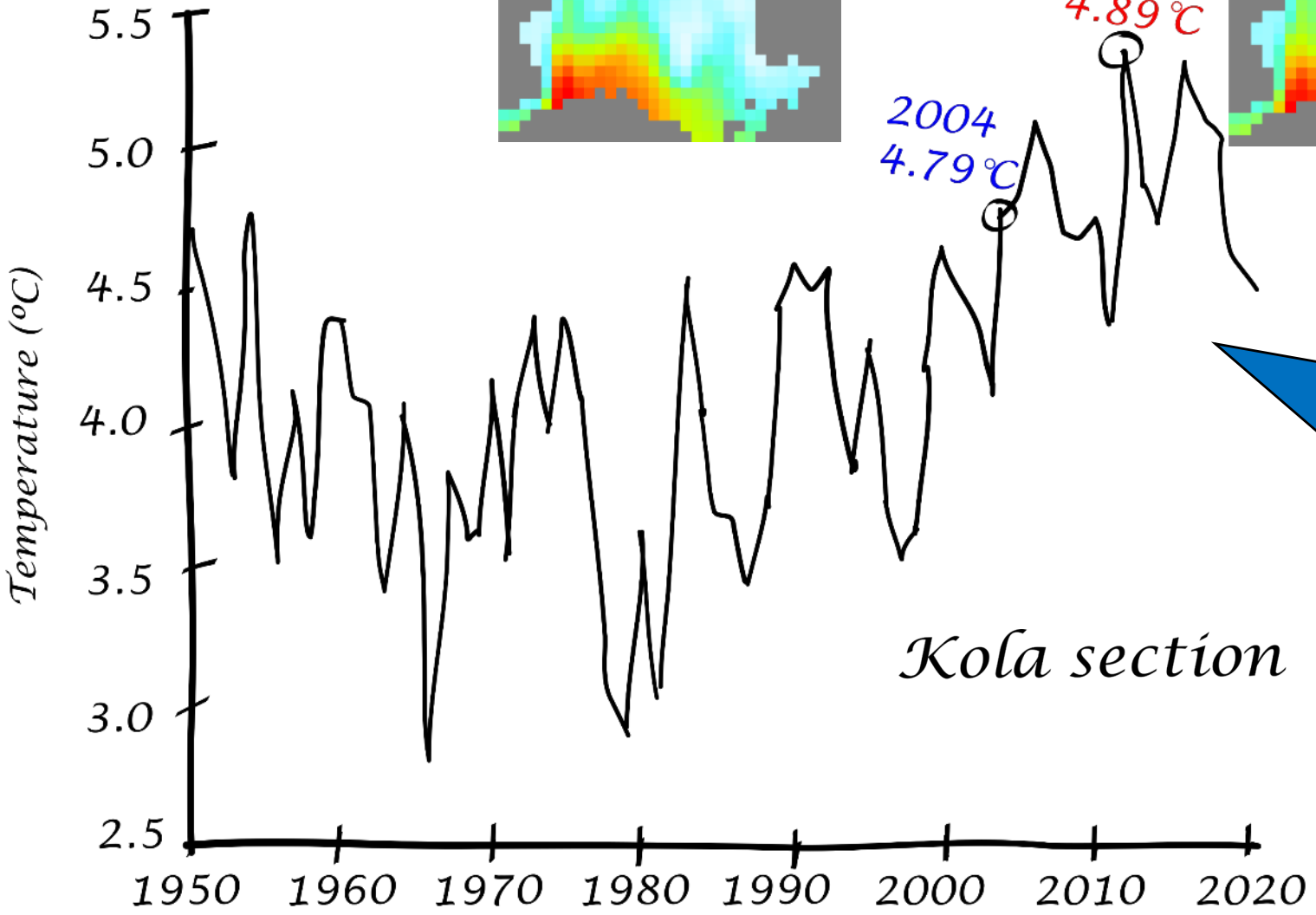
Average 0.89 C

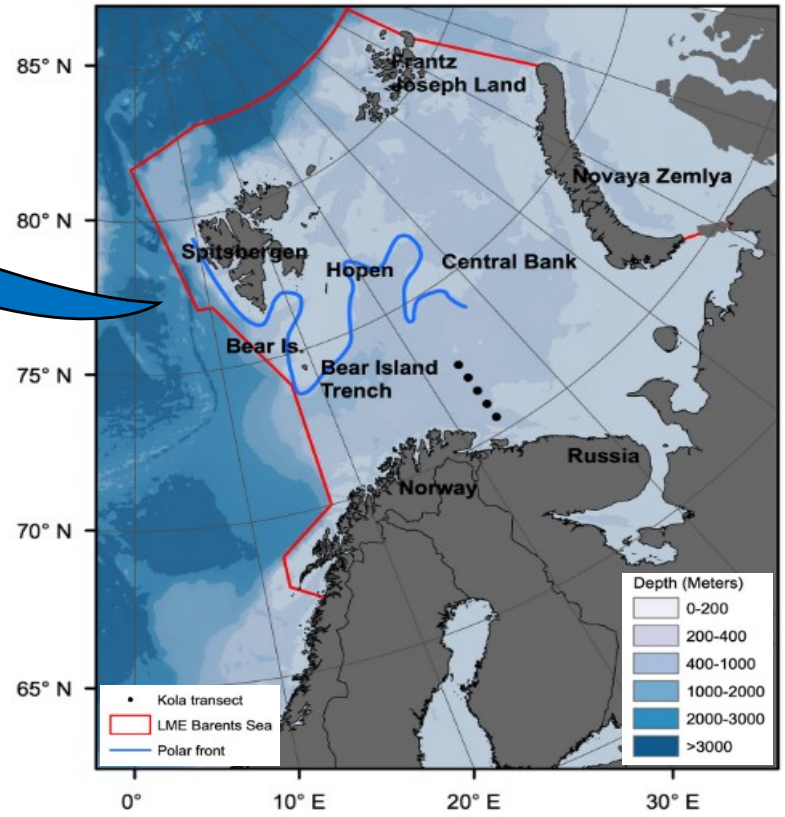


SVIM archives
Regional model
historical data

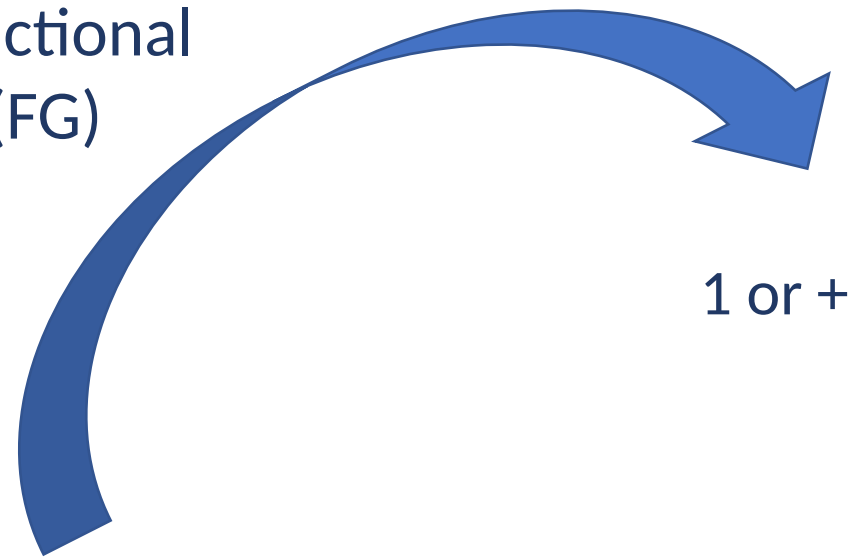
Bottom temperature
Surface temperature
Ice coverage

Primary production





Habitat capacities

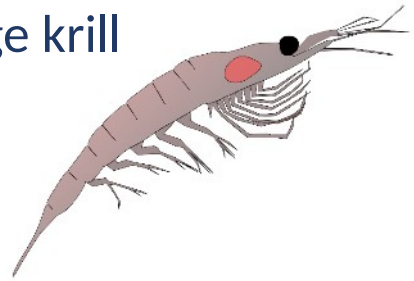


QGAM model (IMR)–ecosystem survey (IMR/PINRO)+ literature

Functional groups with high biomass and good distribution information
35 FGs

Biomass distribution observed vs modelled

Large krill

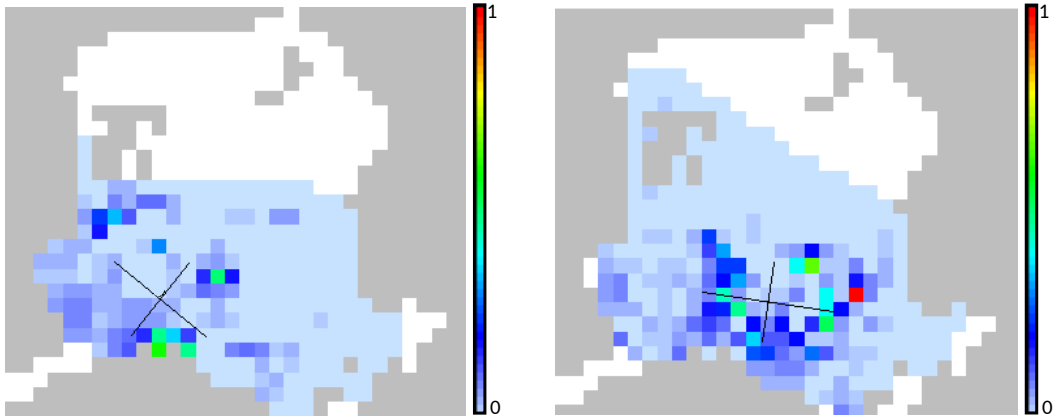


Observed

Cold years
1975 - 2015

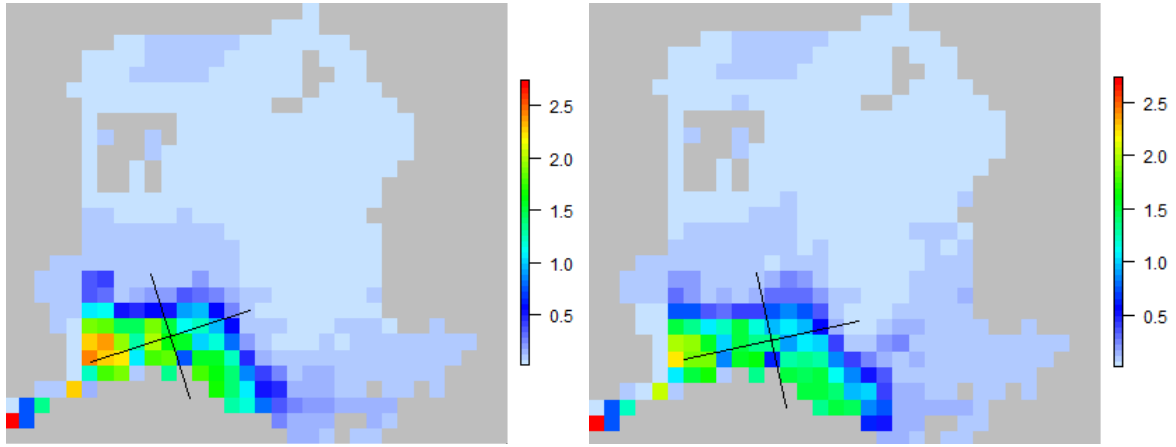
Warm years
1975 - 2015

From literature



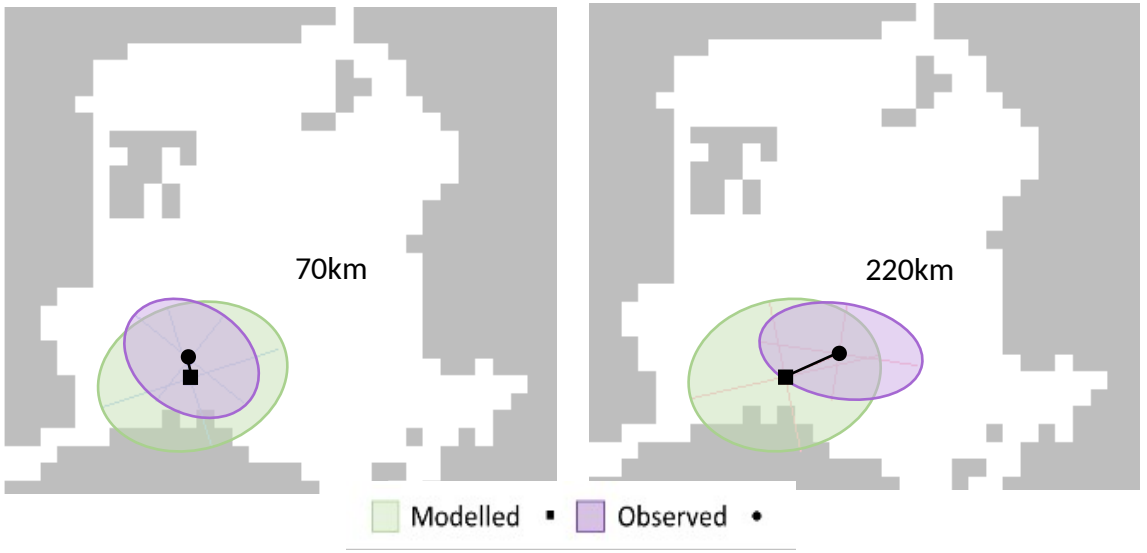
Mikhina et al. (2019), Eriksen and Dalpadado (2011) and Orlova et al. (2010)

Modelled



Cold year

Warm year

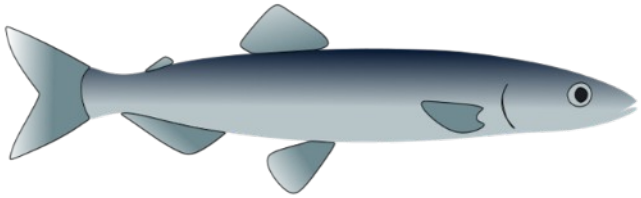


Average diference CG of all FG

Cold year **19** km

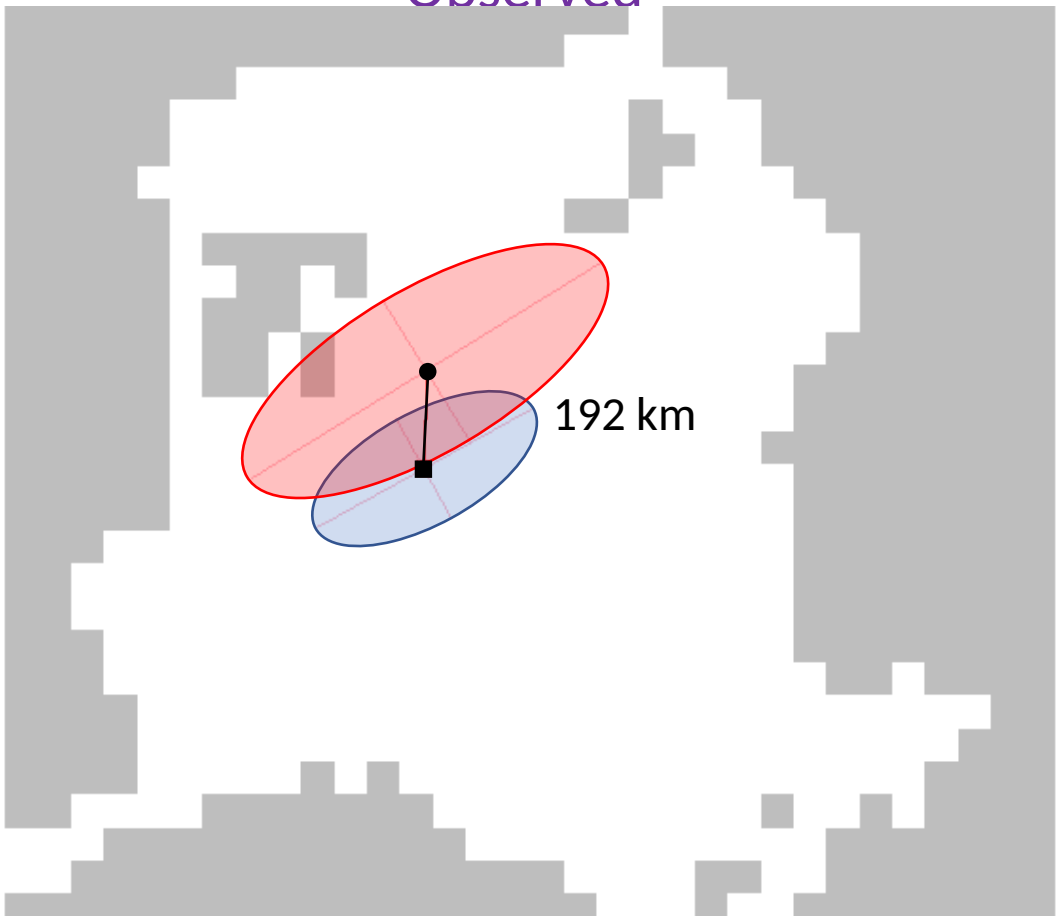
Warm year **26** km

Shift in distributions from cold to warm year

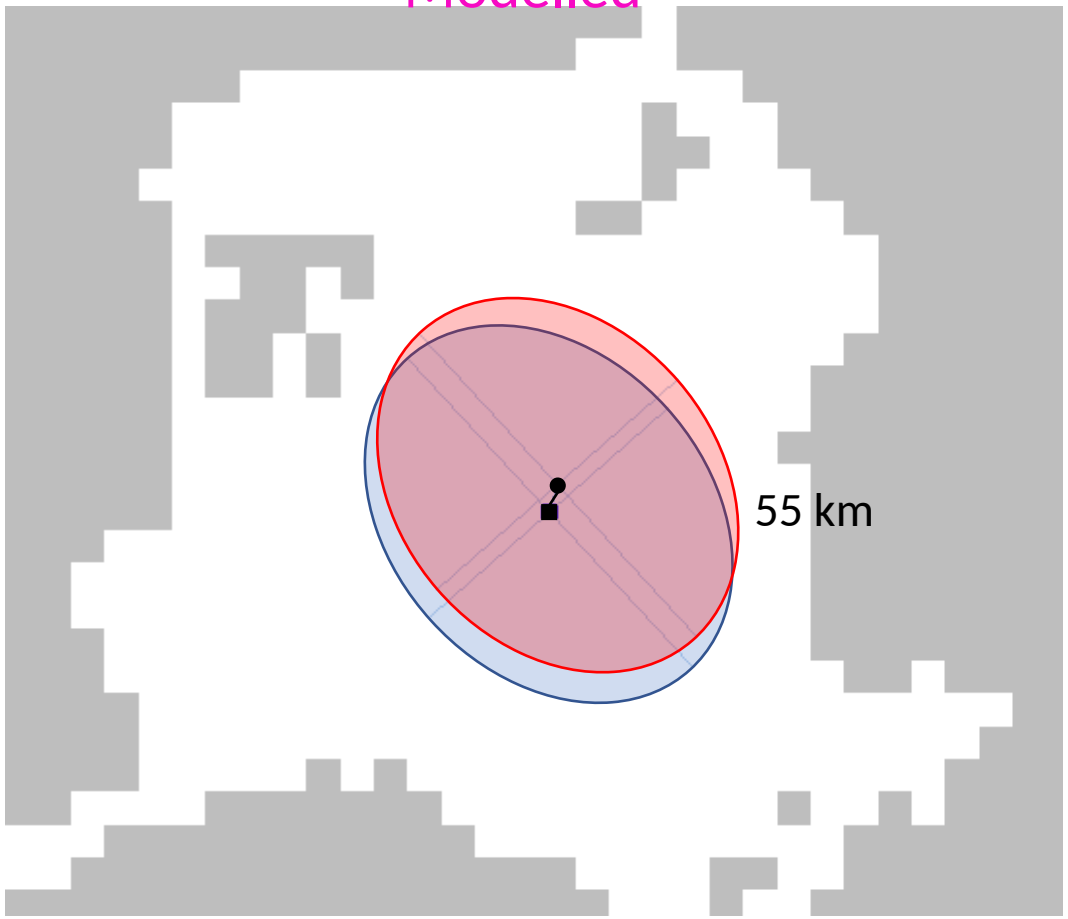


Capelin (3+)

Observed

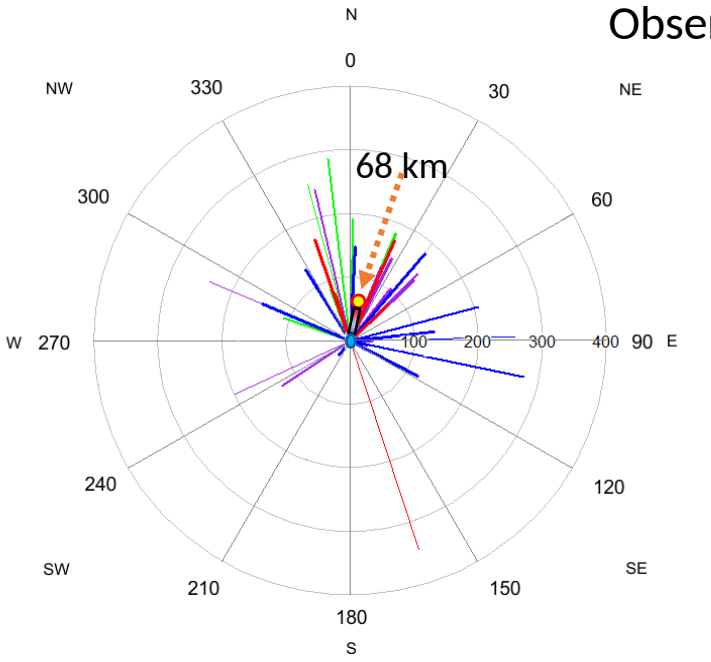


Modelled

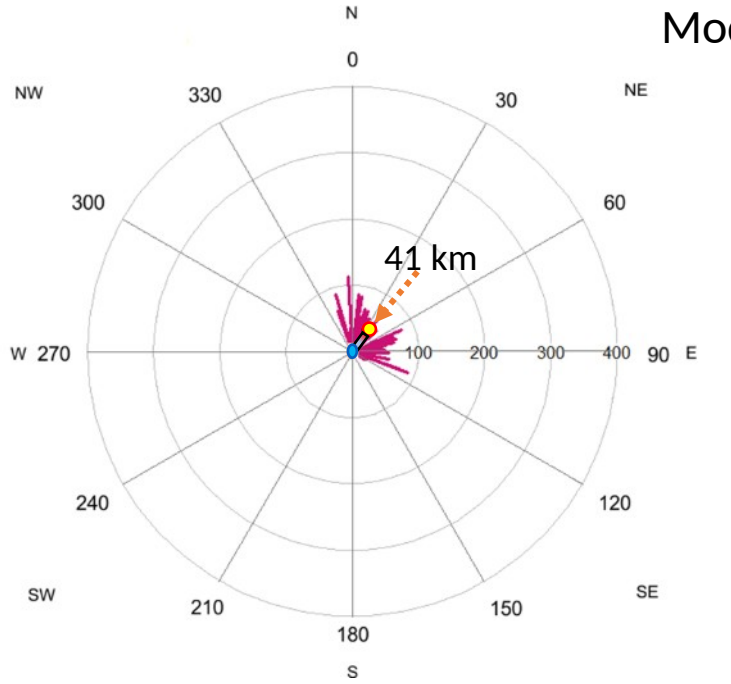


Shift in distributions from cold to warm year

Observed

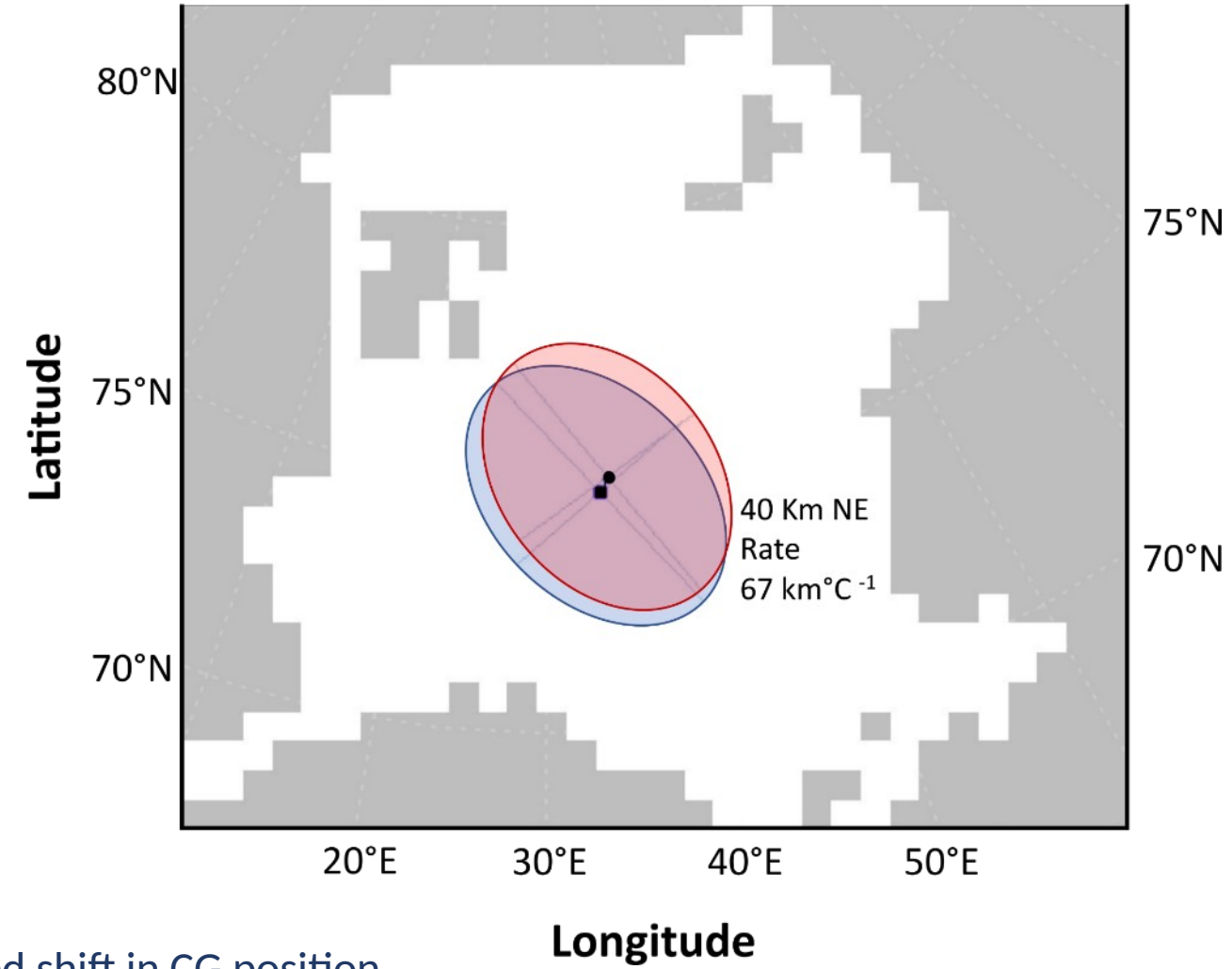


Modelled



35 FGs

Whole Community



Average directed shift in CG position

Summarizing

Q1 How well can a spatial dynamic model predict spatial distributions of functional groups in the last two decades?

The model can represent well the FGs distributions.

Q2 Can the model predict the changes in observed spatial distribution of the functional groups from cold to warm years?

The model can predict well the direction of shift, mainly north-eastern

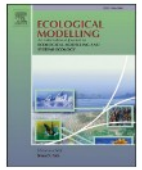
The CG average positions, the displacement direction and the distance predicted by the model are very similar to the survey based observations.

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Modelling the spatial shifts of functional groups in the Barents Sea using a climate-driven spatial food web model

Marcela C. Nascimento [a,*], Berengere Husson [b], Lilia Guillet [b], Torstein Pedersen [a]

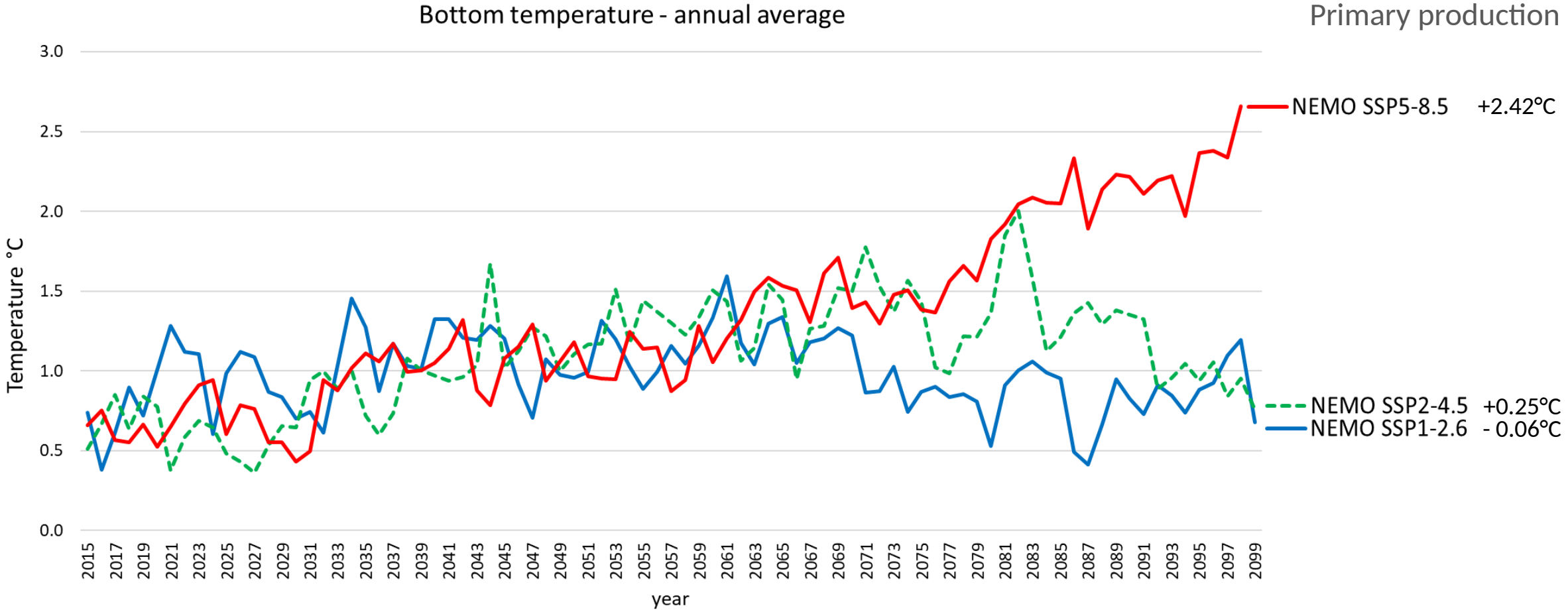
What's now?

Future climate changes effects on spatial distributions

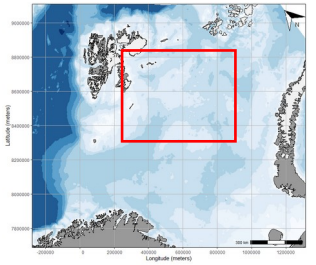
Future scenarios

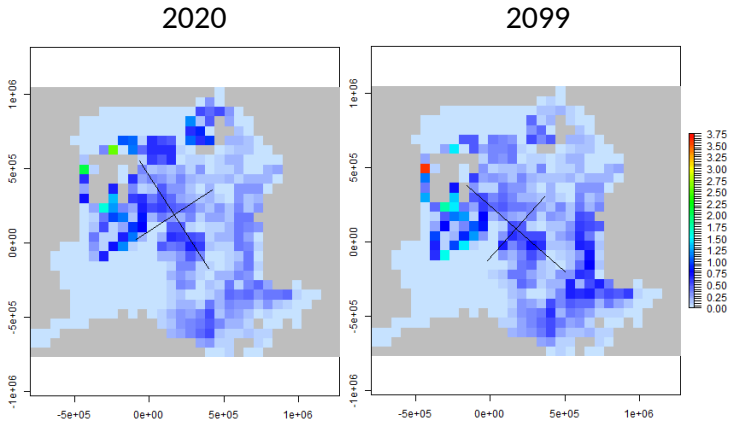
NEMO - NORECOM
(IMR and UIB)

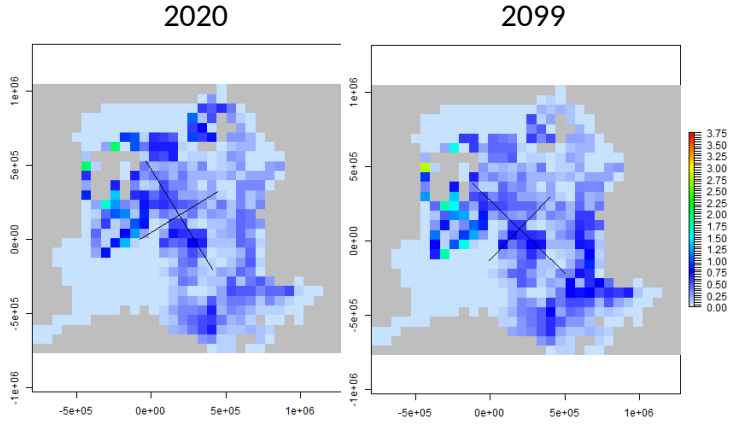
Surface temperature
Ice coverage
Primary production

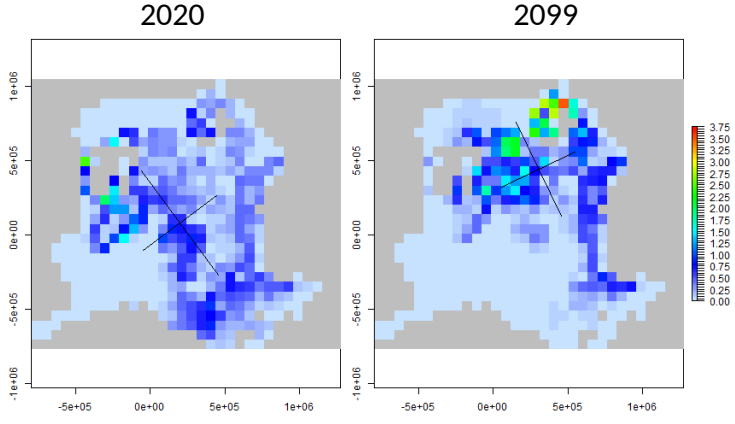


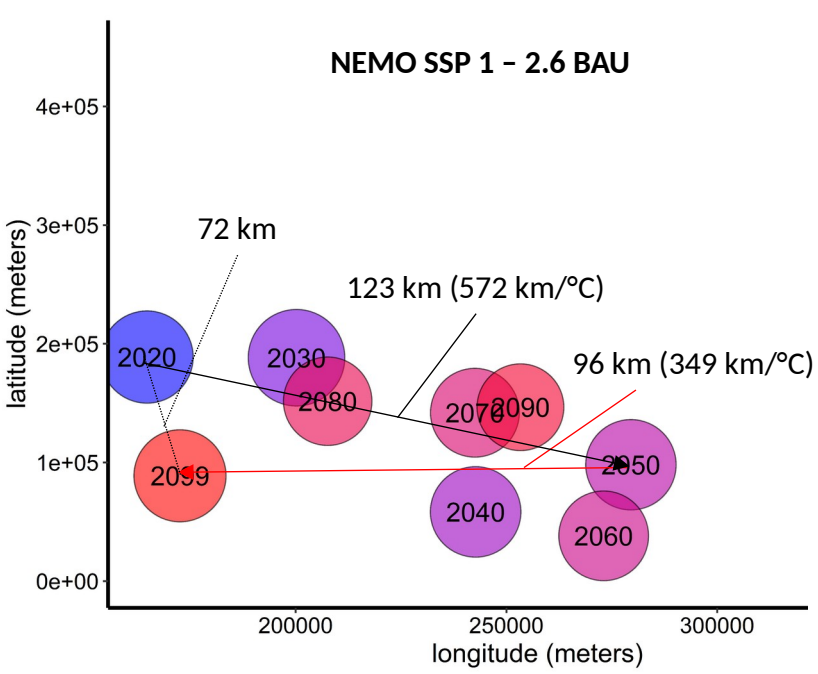
Capelin 0-2

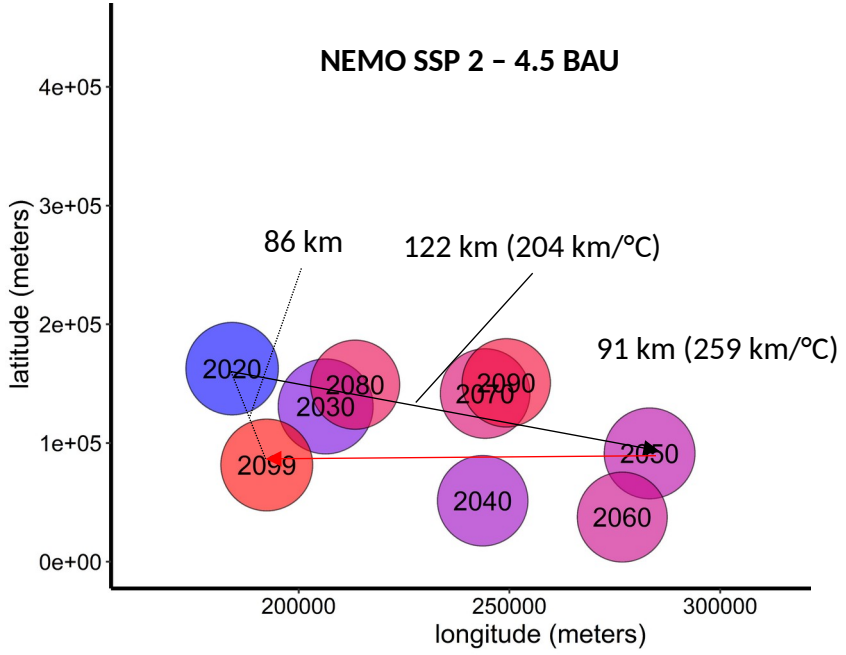


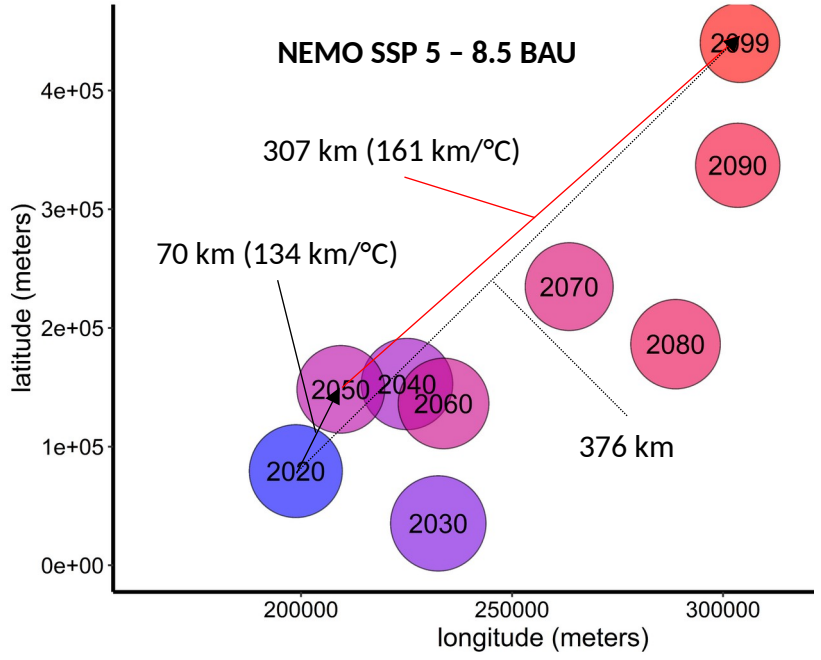

2020
2099


2020
2099

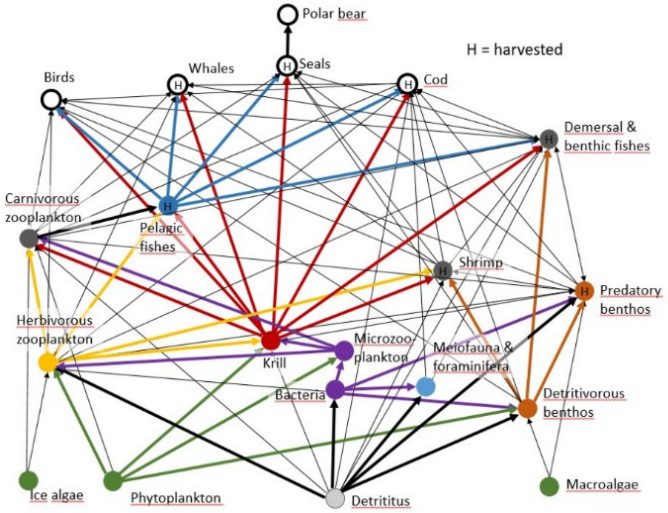

2020
2099

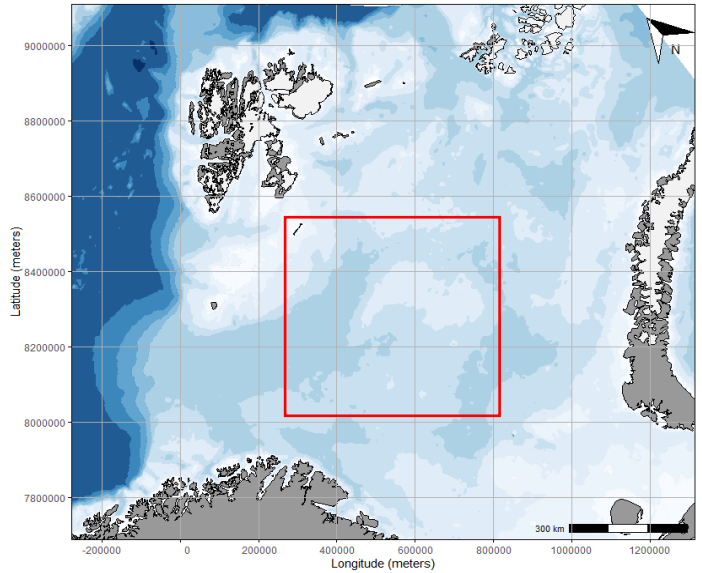

NEMO SSP 1 – 2.6 BAU
72 km
123 km (572 km/°C)
96 km (349 km/°C)
2020
2030
2080
2070
2090
2099
2050
2040
2060
latitude (meters)
longitude (meters)

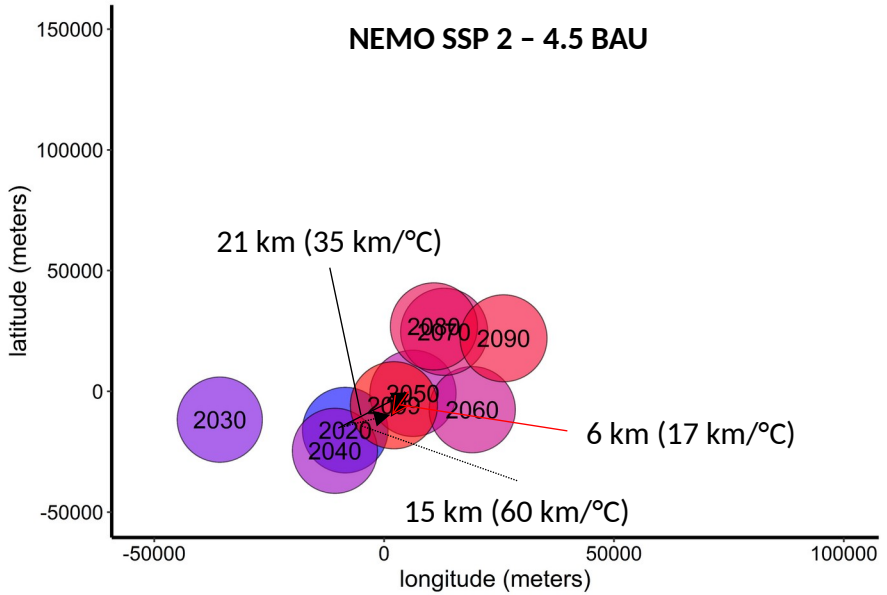

NEMO SSP 2 – 4.5 BAU
86 km
122 km (204 km/°C)
91 km (259 km/°C)
2020
2080
2030
2090
2070
2099
2050
2040
2060
latitude (meters)
longitude (meters)

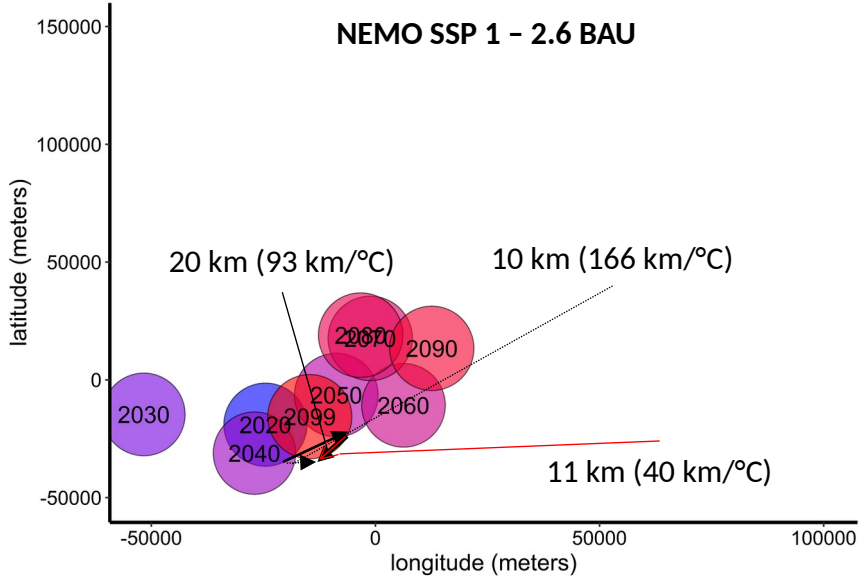

NEMO SSP 5 – 8.5 BAU
307 km (161 km/°C)
70 km (134 km/°C)
376 km
2099
2090
2070
2080
2050
2040
2060
2020
2030
latitude (meters)
longitude (meters)

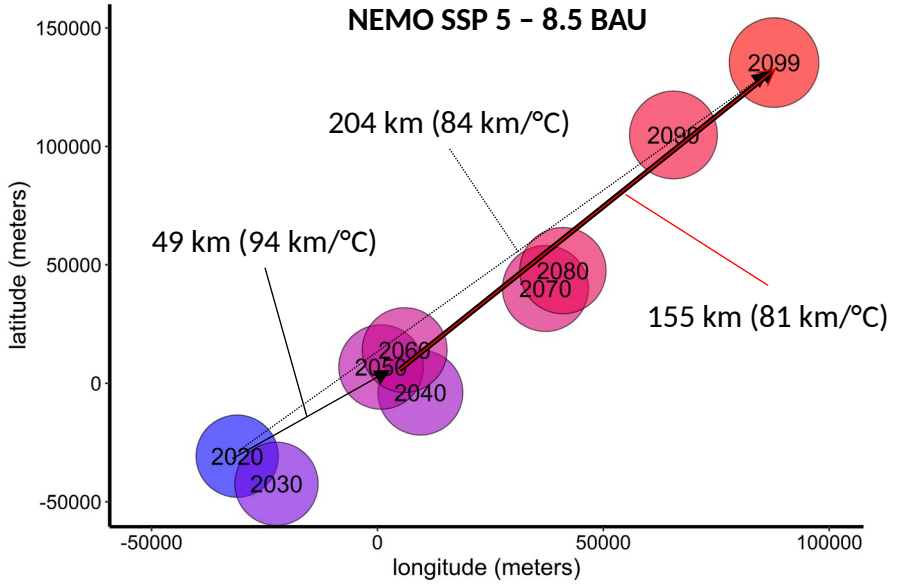
Whole community











Thank you



Common Gull *Larus canus*
FG 21 (Other gulls and surface feeder)